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Experiment: Deploy a Multi-Service Application Using Docker Compose

Aim

To deploy and manage a multi-service application using Docker Compose by creating and running multiple containers from a single configuration file.

Objective

- To understand Docker Compose.
- To deploy multiple services using a single `docker-compose.yml` file.
- To build and run multiple containers together.
- To enable communication between different containers.

This Experiment maps t**o LO4**

LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose,

Kubernetes, and cloud infrastructure tools.

Theory

Docker is a containerization platform that packages applications along with their dependencies into lightweight containers. Containers provide a consistent environment across development, testing, and production.

Docker Compose is a tool used to define and manage multi-container Docker applications using a single configuration file (`docker-compose.yml`). It automatically creates the required containers, networks, and volumes, allowing all services to be deployed with one command.

In this experiment, the application consists of:

- **Backend:** Flask Web Application
- **PostgreSQL:** Database Service

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- **Redis:** Cache Service

Docker Compose creates an internal network so that all services can communicate using their service names

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Procedure

1. Create the project directory.
2. Create the backend application files.
3. Create the `Dockerfile`.
4. Create the `docker-compose.yml` file.

```yaml
version: "3.9"



services:

  backend:

    build: ./backend

    ports:

      - "5000:5000"

    depends_on:

      - postgres

      - redis



  postgres:

    image: postgres:16

    environment:

      POSTGRES_USER: admin
```

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```
      POSTGRES_PASSWORD: password

      POSTGRES_DB: mydb

    ports:

      - "5432:5432"



  redis:

    image: redis:7

    ports:

      - "6379:6379"
```

5. Build the Docker images.
6. Start all services using Docker Compose.
7. Verify the running containers.
8. View application logs.
9. Stop and remove the containers.

Command	Description	Example Usage
`docker compose up`	Creates and starts all services defined in `docker-compose.yml`. It creates containers, networks, and volumes if they do not exist.	`docker compose up`
`docker compose up -d`	Starts services in **detached mode** (runs containers in the background).	`docker compose up -d`
`docker compose down`	Stops and removes containers and networks created by Docker Compose.	`docker compose down`

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`docker compose down -v`	Stops containers and removes containers, networks, and attached volumes. Useful for complete cleanup.	`docker compose down -v`
`docker compose build`	Builds Docker images for services defined with the `build` option in `docker-compose.yml`.	`docker compose build`
`docker compose up --build`	Rebuilds images and starts containers in one command.	`docker compose up --build`
`docker compose start`	Starts existing stopped containers without recreating them.	`docker compose start`
`docker compose stop`	Stops running containers without removing them.	`docker compose stop`
`docker compose restart`	Restarts all running services.	`docker compose restart`

Implementation

Step 1: Build the Docker Images

Command

docker compose build

Output

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```
C:\Users\15L\multi-service-app>docker compose build
time="2026-07-29T13:54:49+05:30" level=warning msg="C:\\Users\\15L\\multi-service-app\\docker-compose.
yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
#1 [internal] load local bake definitions
#1 reading from stdin 561B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 197B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11-slim
#3 DONE 1.7s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 93B done
#5 DONE 0.0s

#6 [1/5] FROM docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c
7627accd3d51053a93
#6 resolve docker.io/library/python:3.11-slim@sha256:db3ff2e1800a8581e2c48a27c3995339d47bdf046da21c762
7accd3d51053a93 0.0s done
#6 DONE 0.0s

#7 [2/5] WORKDIR /app
#7 CACHED

#8 [3/5] COPY requirements.txt .
#8 CACHED

#9 [4/5] RUN pip install -r requirements.txt
#9 CACHED

#10 [5/5] COPY . .
#10 CACHED

#11 exporting to image
#11 exporting layers done
#11 exporting manifest sha256:489dcdc05bef1f4e5aa358e94fa9fcd75c331ff13ca2bab6f514450b13a19c8a done
#11 exporting config sha256:39c28977e23353c413866a304add1861f41fab4c8df3994555c0760279aef484 done
#11 exporting attestation manifest sha256:43dbad46a3a8624e42749d8d25543f8062b571a9097af573be9b42f24b17
e1b2 0.0s done
#11 exporting manifest list sha256:dfdb5b94c3bec130bc2992a8bf856a1e57b9114d41af45e13372621ed597dd07 do
ne
#11 naming to docker.io/library/multi-service-app-backend:latest done
```

Step 2: Start All Services

Command

docker compose up -d

Output

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```
C:\Users\15L\multi-service-app>docker compose up -d
time="2026-07-29T13:55:23+05:30" level=warning msg="C:\\Users\\15L\\multi-service-app\\docker-compose.
yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
[+] up 2/2
 ✔Container multi-service-app-postgres-1 Running                                                  0.0s
 ✔Container multi-service-app-backend-1  Recreated                                                1.4s

C:\Users\15L\multi-service-app>
```

Step 3: Verify Running Containers

Command

docker compose ps

Output

```
C:\Users\15L\multi-service-app>docker compose ps
time="2026-07-29T13:55:49+05:30" level=warning msg="C:\\Users\\15L\\multi-service-app\\docker-compose.
yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
NAME                           IMAGE                      COMMAND                  SERVICE    CREATED
              STATUS          PORTS
multi-service-app-backend-1    multi-service-app-backend   "python app.py"          backend    About a
 minute ago   Up 24 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
multi-service-app-postgres-1   postgres:16                 "docker-entrypoint.s…"   postgres   27 minu
tes ago       Up 27 minutes   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
multi-service-app-redis-1      redis:7                     "docker-entrypoint.s…"   redis      24 minu
tes ago       Up 24 seconds   6379/tcp

C:\Users\15L\multi-service-app>
```

Step 4: View Container Logs

Command

docker compose logs

Output

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```
C:\Users\15L\multi-service-app>docker compose logs
time="2026-07-29T13:56:22+05:30" level=warning msg="C:\\Users\\15L\\multi-service-app\\docker-compose.
yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
postgres-1  | The files belonging to this database system will be owned by user "postgres".
postgres-1  | This user must also own the server process.
postgres-1  |
postgres-1  | The database cluster will be initialized with locale "en_US.utf8".
postgres-1  | The default database encoding has accordingly been set to "UTF8".
postgres-1  | The default text search configuration will be set to "english".
postgres-1  |
postgres-1  | Data page checksums are disabled.
postgres-1  |
postgres-1  | fixing permissions on existing directory /var/lib/postgresql/data ... ok
postgres-1  | creating subdirectories ... ok
postgres-1  | selecting dynamic shared memory implementation ... posix
postgres-1  | selecting default max_connections ... 100
postgres-1  | selecting default shared_buffers ... 128MB
postgres-1  | selecting default time zone ... Etc/UTC
postgres-1  | creating configuration files ... ok
postgres-1  | running bootstrap script ... ok
postgres-1  | performing post-bootstrap initialization ... ok
postgres-1  | syncing data to disk ... ok
backend-1   |  * Serving Flask app 'app'
backend-1   |  * Debug mode: off
```

Step 5: Stop Running Containers

Command

docker compose stop

Output

```
C:\Users\15L\multi-service-app>docker compose stop
time="2026-07-29T13:56:58+05:30" level=warning msg="C:\\Users\\15L\\multi-service-app\\docker-compose.
yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
[+] stop 3/3
 ✔Container multi-service-app-backend-1  Stopped                                    1.2s
 ✔Container multi-service-app-redis-1    Stopped                                    0.1s
 ✔Container multi-service-app-postgres-1 Stopped                                    0.3s

C:\Users\15L\multi-service-app>
```

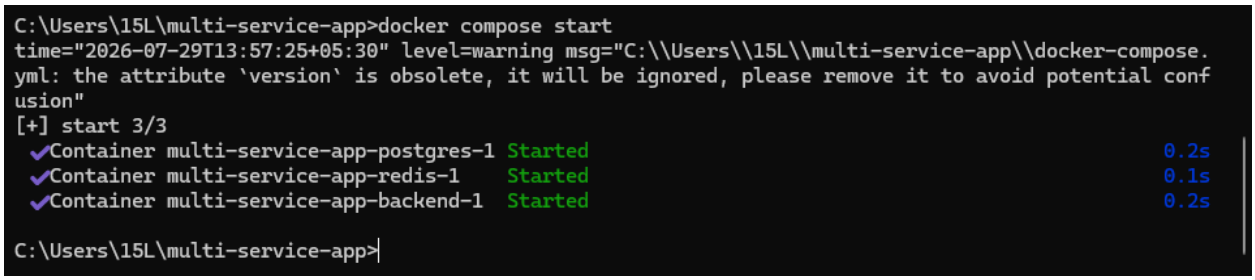
Step 6: Start Stopped Containers

Command

docker compose start

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Output

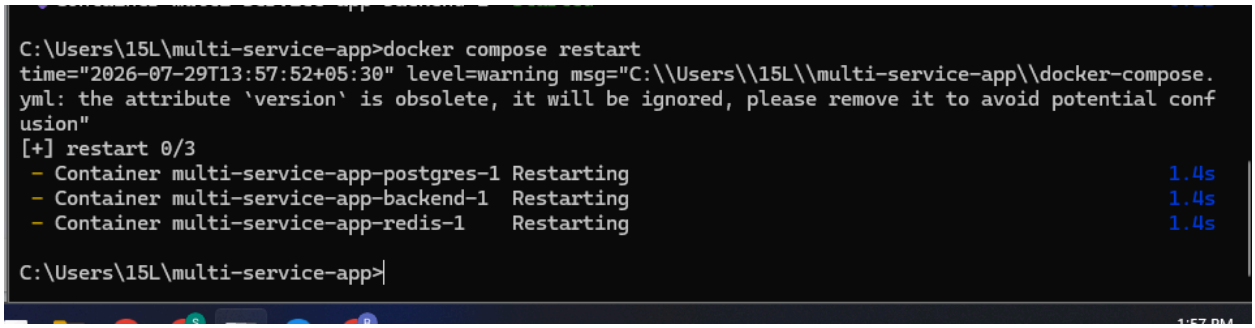


Step 7: Restart Containers

Command

docker compose restart

Output



Step 8: Remove Containers and Volumes

Command

docker compose down -v

Output

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```
C:\Users\15L\multi-service-app>docker compose down -v
time="2026-07-29T13:58:27+05:30" level=warning msg="C:\\Users\\15L\\multi-service-app\\docker-compose.
yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential conf
usion"
[+] down 4/4
 ✔Container multi-service-app-backend-1  Removed                                    1.3s
 ✔Container multi-service-app-redis-1    Removed                                    0.2s
 ✔Container multi-service-app-postgres-1 Removed                                    0.4s
 ✔Network multi-service-app_default      Removed                                    0.3s

C:\Users\15L\multi-service-app>
```

Result

The multi-service application was successfully deployed using Docker Compose. The Flask backend, PostgreSQL database, and Redis cache were configured and managed through a single `docker-compose.yml` file, demonstrating efficient deployment and orchestration of multiple Docker containers.